Radial polarising dome, final cutting guide, axis confirmed

R = 40 mm, n = 8 gores, PiCam360, one sheet cut of four.

TRANSMISSION AXIS: CONFIRMED PARALLEL TO THE LONG (20 cm) EDGE OF THE SHEET

Three independent tests agree: outdoor sky against sunlight, the Zephyrus G14 laptop display, and a polarised-sunglasses cross-check with a known vertical reference axis. All three go dark in landscape orientation, meaning the axis runs the long way across the sheet. There is no remaining ambiguity and the earlier two readings (A versus B) are resolved in favour of A. Cut directly from page 2, no further testing needed.

What this means for the template

The gore centre line, which is the transmission axis on the physical film, must run parallel to the sheet's long edge. Practically, this means the gore's long dimension, apex to base, lies along the 20 cm side of the sheet, and the template is laid down square to the sheet with no rotation. This is Layout A from the earlier draft, now the only layout, since the axis question that made Layout B a live option is closed.

Locked geometry

dome radius R	**40.0 mm**
gores n	**8**
gore length, apex to base ($L = \pi R/2$)	**62.83 mm**
gore width at base ($W = 2\pi R/n$)	**31.42 mm**
base circumference ($n \times W = 2\pi R$)	**251.3 mm**
octagon lid, circumradius	**40.0 mm**
octagon lid, edge length	**30.61 mm**
octagon lid, inradius	**36.96 mm**
PiCam360 half diagonal (~38×38 mm board)	**26.87 mm (clears inradius)**
tuck tab	**8 mm × 28 mm, folds 90°**
sheets required	**1 of 4 (3 spare)**

Mount

The UTU shipping box lid, cut with a regular octagon and eight edge slits, each gore's tab folded 90° and pushed into its slit. No adhesive on the base ring, so mistakes stay reversible until the seams are taped. Full pattern and slit dimensions on page 3.

Record before cutting, write these numbers down now

date and time of the sunglasses confirmation test: ____________________

which physical edge of the sheet the axis line was drawn along (long edge, confirmed above): long edge

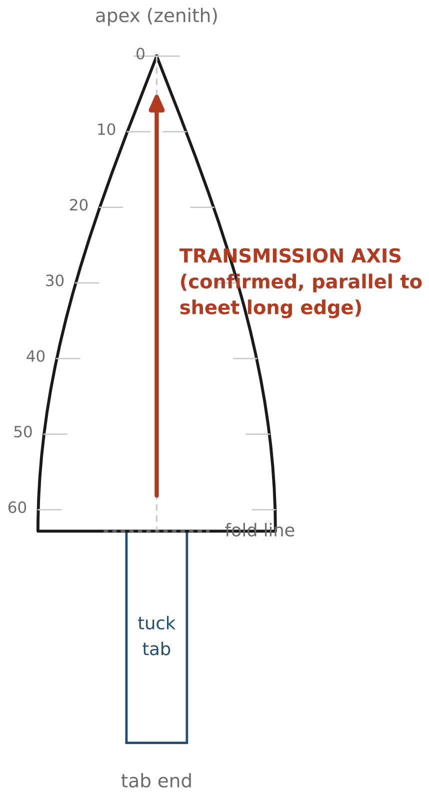
measured dome radius after gores are cut and joined: ______ mm (design value 40.0 mm)

octagon lid measured circumradius after cutting: ______ mm (design value 40.0 mm)

Master gore template, printed at 1:1

Print at 100 percent / Actual size, page scaling and fit-to-page OFF. Axis arrow = confirmed transmission axis.

SCALE CHECK: this bar must measure exactly 100 mm. If it does not, reprint at 100 percent.



Key dimensions

dome radius R	40.0 mm
gores n	8
gore length, apex to base	62.83 mm
gore width at base	31.42 mm
tuck tab	8 x 28 mm
sheets required	1 (of 4)

Half width at arc distance s from the apex

half width = 15.708 x sin(s / 40) with s and the result in mm

s	half	full
0.00	0.00	0.00
5.00	1.96	3.92
10.00	3.89	7.77
15.00	5.75	11.51
20.00	7.53	15.06
25.00	9.19	18.38
30.00	10.71	21.41
35.00	12.06	24.11
40.00	13.22	26.44
45.00	14.17	28.35
50.00	14.91	29.81
55.00	15.41	30.82
60.00	15.67	31.34
62.83	15.71	31.42

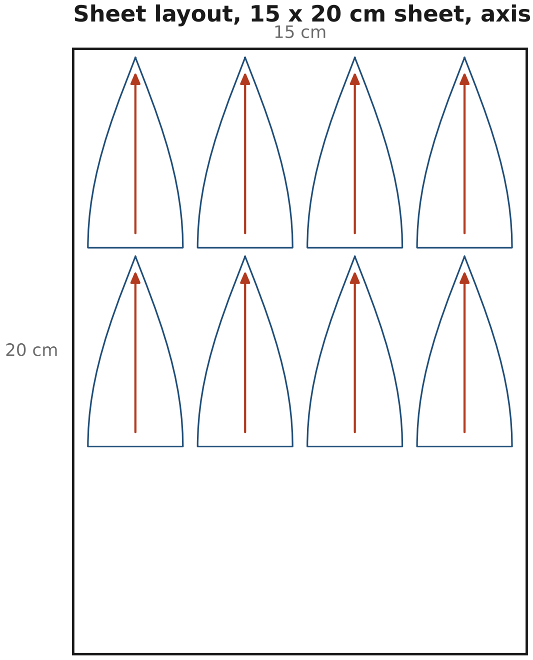
If the print scales wrongly, draw the template by hand from this table.

Cut 8 of these, all from the axis orientation confirmed on page 1: gore length along the sheet's long (20 cm) edge. Trace all 8 first, mark the axis arrow on the back of each before cutting, then cut slowly. Number the gores 1 to 8 on the back so the assembly order is recoverable.

Sheet layout and octagon lid mount, printed at 1:1 (octagon only)

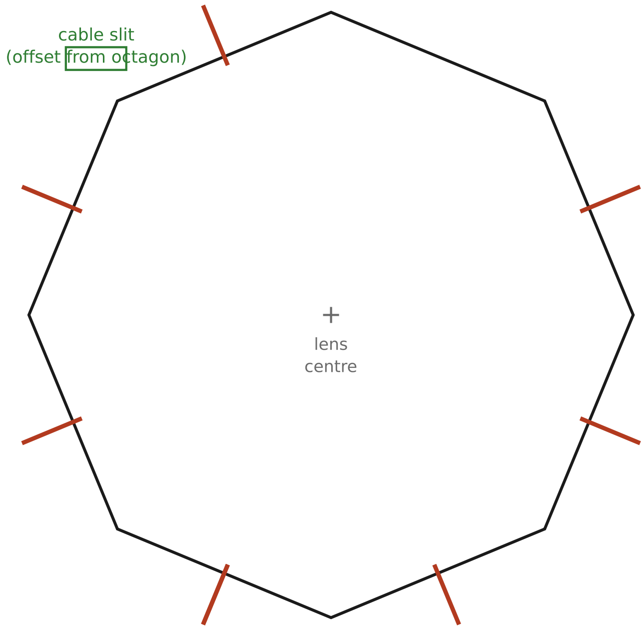
One layout only, since the axis is confirmed. Assembly steps continue on page 4.

Sheet layout, 15 x 20 cm sheet, axis along the 20 cm edge



4 across x 3 down = 12 possible, cut 8, 4 spare on this sheet

Octagon lid, circumradius 40 mm (= R)



edge 30.61 mm, inradius 36.96 mm
red = 8 mm slit at each edge midpoint

The octagon above is printed at 1:1 and is the actual template for the box lid: cut along the eight sides and mark the eight red slit positions directly from this page. The sheet layout is a small arrangement diagram only, not a cutting template; use the 1:1 gore template on page 2 for the actual film cutting.

Assembly, mounting and what changes in the draft

Final page. Read fully before starting; step 6 (levelling and marking North) cannot be redone later.

Assembly

1. Butt the long curved edges of gore 1 and gore 2 together, edge to edge, not overlapped, since an overlap repeated eight times shrinks the base circumference below the 40 mm design radius. Bridge each joint with a narrow strip of tape on the outside only, no wider than about 5 mm.
2. Continue 2 to 3, 3 to 4, through to 8, then close 8 back to 1.
3. Gather the eight apex tips and fix them together at the top with a small piece of tape.
4. On the box lid, draw the octagon at 40 mm circumradius, centred on the lens axis, and cut the eight 8 mm slits at the edge midpoints, copied from the 1:1 template on page 3. Cut the separate cable slit, offset from the octagon.
5. Fold each gore's tab 90 degrees along its fold line and push it through its slit from above. No adhesive on the base ring; the tab and slit alone hold the dome open and in shape.
6. Level the lid and mark North on it before anything else is fixed down, since every heading estimate from this point onward is measured relative to that mark.

Tape caution

Most clear tapes are birefringent and rotate polarisation. Keep tape strictly on the seams and the apex, never across the open area of a gore. The eight seams will show as faint radial lines in captured frames; mask a couple of degrees either side of each in processing and record that you did so.

After mounting, before any real capture

Take one test frame. Confirm the lid material does not rise into the fisheye's field of view; if it does, it will appear as a dark intrusion at the horizon in every frame captured afterward, and is much easier to fix now, by trimming the lid edge, than to notice later in a full capture session.

Changes this confirms in the draft

Section 4.2 should read R = 40 mm, N = 8, gores 62.83 mm long and 31.42 mm wide at the base, all eight cut from a single sheet, mounted on an octagonal box-lid base with tuck tabs rather than a taped card ring. Record the as-built radius once assembled, alongside the 40 mm design figure.